

Topology Behind Jujutsu Kaisen: Ryomen Sukuna's Cuts



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1 Introduction

Everyone knows what it means to cut something. You know it so well that the question of what cutting actually *is* never comes up, which is a shame, because the answer took mathematics until 1887 to state properly and another twenty-five years to prove.

Ryomen Sukuna cuts things for a living. His innate technique, Shrine, does nothing else. What makes him worth an essay is that over the course of the series he cuts three different kinds of object, and the escalation is not one of power. It is one of category. Dismantle cuts a thing. Cleave cuts a thing while reading how tough it is. And in chapter 236, against an opponent whose entire defence was built on distance, Sukuna stopped cutting things inside space and cut space.

That last move killed Gojo Satoru, and the manga's explanation of why is three words long: he broadened the target.

This essay is about what those three words buy you. The tools are topology, which studies the properties of a space that survive bending and stretching, and differential geometry, which studies the ones that do not. Between them they say something precise about why a slash works, why it is astonishingly cheap, and why Infinity had no answer to the last one.

Four results. The number of pieces a body comes apart into grows as the *cube* of the number of slashes, so ten cuts make a hundred and seventy-six fragments. Cutting a skyscraper costs roughly a hundred thousand times less energy than the collapse it sets off, because a cut pays for area while destruction pays for volume. There is a single property, codimension, that decides whether a cut separates anything at all, and it explains why Sukuna's technique is slashes rather than stabs. And Gojo's Infinity and Sukuna's final slash are both sets of measure zero, which sounds like it should make them comparable and does not, for a reason that measure theory is constitutionally unable to see.

2 What the Three Cuts Actually Do

Shrine (御廚子, *Mizushi*) debuts in chapter 8. It lets Sukuna produce invisible slashes at range or on contact, and the manga splits it into two attacks with an explicit selection rule stated in chapter 119: Dismantle strikes targets without cursed energy, Cleave strikes targets with it.

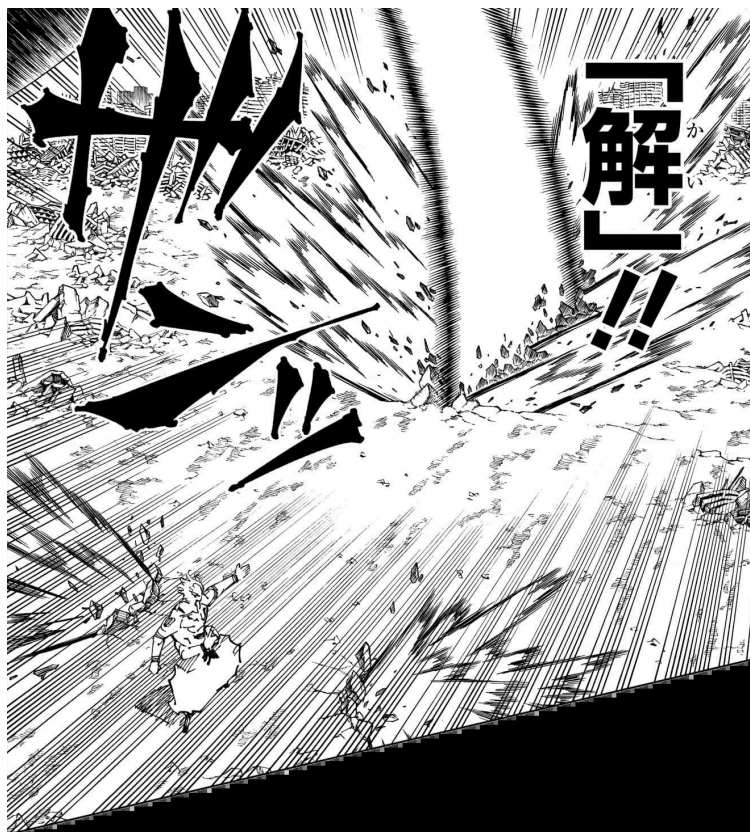


Figure 1: Dismantle. The slash itself is invisible and arrives at range; what the art shows is the aftermath, which is the only part of the technique that has ever been visible. Image source: Jujutsu Kaisen Wiki (Fandom), *Shrine*, retrieved September 2026. Original work by Gege Akutami, Shueisha.

Dismantle (解, *Kai*) is the default. Its power scales with cursed energy output, and canon brackets that range at both ends. At under ten percent output a barrage of it left Yuji Itadori with bloody gashes and nothing worse (chapters 214 and 215). At full output, one slash cut a skyscraper in half (chapter 224). Mahoraga eventually learned to see and deflect it, and Maki Zenin dodged one on reflexes alone, so it is fast rather than instantaneous.

Cleave (捌, *Hachi*) is the more interesting of the two. The manga describes it as adjusting its power according to the target's toughness and cursed energy level, so as to cut them down in a single attack. Outside a domain it requires physical contact. The version applied to the environment, Spiderweb (蜘蛛の糸, *Kumo no Ito*), adjusts to the toughness of the surroundings, carves the ground into a web pattern, and collapses the whole thing at once (chapter 215).

Malevolent Shrine, the domain, runs both continuously inside a sure-hit region that Sukuna can push out to a radius of nearly two hundred metres (chapter 119). That is a hemisphere of about seventeen million cubic metres, being slashed without pause.

Then chapter 236. Using what Mahoraga had worked out about Limitless, Sukuna enhanced Dismantle by broadening its target area to include space, existence and the world themselves, producing what the manga names a slash that cuts the world (世界

を断つ斬撃, *Sekai o Tatsu Zangeki*). It went through Infinity, a barrier that had never been penetrated by anything, and killed Gojo by slicing him in half.

Every word of that is a claim about what the slash is applied *to* rather than how hard it is applied. Hold onto that.

3 What It Means to Cut Something

Take a solid object and think of it as a set of points. Before any slash, that set is connected: you can travel from any point of it to any other without leaving it. After a clean slash, you cannot. The set has come apart into two pieces, and the technical name for a piece is a connected component.

Topologists count components with a number written b_0 , the zeroth Betti number. A single lump has $b_0 = 1$. Two lumps have $b_0 = 2$. Cutting, in the only sense that matters, is any operation that raises b_0 .

This sounds too simple to need a theorem, and that impression is exactly why the subject is interesting. The statement that a closed loop drawn in the plane divides it into an inside and an outside is called the Jordan curve theorem. Camille Jordan wrote it down in 1887, and mathematicians then spent a century arguing about whether his proof was sound, with Veblen supplying a replacement in 1905 and Hales arguing in 2007 that Jordan had been right all along. Brouwer generalised it in 1912: a compact connected hypersurface embedded in $\mathbb{R}^n$ divides it into exactly two pieces, and is the boundary of each. Obvious, and hard.

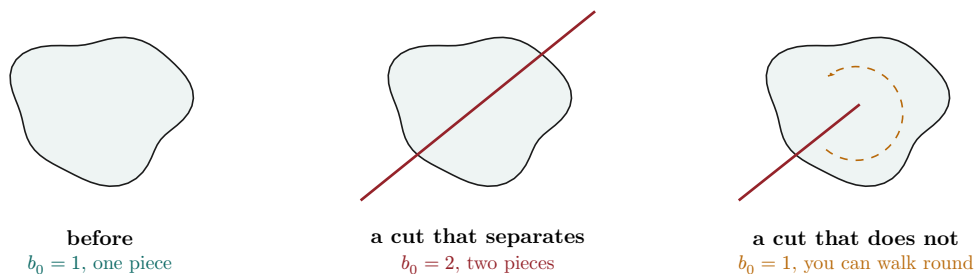


Figure 2: Cutting, defined. A cut is any operation that raises the number of connected components b_0 . The middle blob has been separated and the right one has not, even though the right one has been damaged in a way that looks similar. Whether a slash separates anything depends on where it ends, not on how hard it was thrown, and the next section turns that observation into a rule.

4 Codimension, or Why Slashes and Not Stabs

Here is a fact that sounds false the first time you hear it. Take ordinary three-dimensional space and delete an infinite straight line from it. What is left is still connected. Any two surviving points can be joined by a path, because you can simply walk around the line.

Delete a plane instead and space falls into two halves that can never reach each other.

Both deleted sets are infinite. Both have zero volume. The difference is dimension, and the quantity that matters is not the dimension of the thing you removed but how much lower it is than the space it sits in. That difference is called the codimension: a plane in three-space has codimension $3 - 2 = 1$, a line has codimension 2, a point codimension 3.

The rule is exact. Only a set of codimension 1 can disconnect the space around it. Anything thinner leaves room to go around.

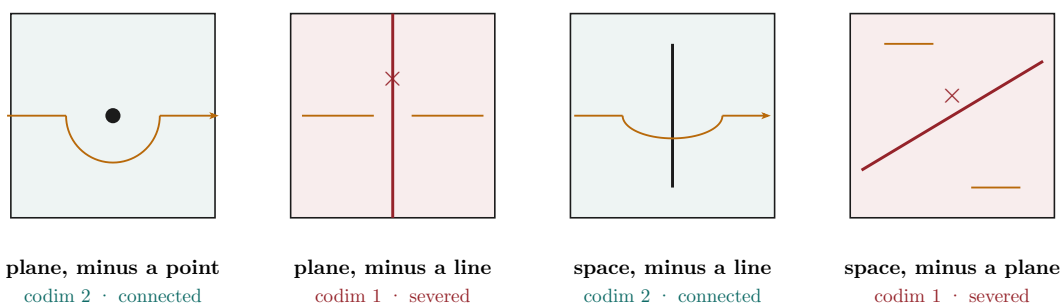


Figure 3: The rule that governs every cut ever made. Removing a set of codimension 2 or more leaves the space connected, because a path can go around it (amber). Only codimension 1 severs. This is why a stab wound is survivable in a way a bisection is not, and why every attack in Sukuna’s arsenal is a slash: in three dimensions, the thinnest thing that can separate anything is a surface.

That rule quietly explains the shape of Sukuna’s entire technique. A stab is a curve swept through space, codimension 2, and it cannot separate anything no matter how deep it goes. A slash is a surface, codimension 1, and separation is the only thing it does. In three dimensions, a two-dimensional sheet is the cheapest object that can divide the world, and Sukuna throws nothing else.

5 Counting the Pieces



Figure 4: The output of a few seconds of Dismantle. The manga is consistent that Sukuna does not saw at things; he lays down a small number of intersecting planes and lets the combinatorics do the work. Image source: Jujutsu Kaisen Wiki (Fandom), *Shrine*, retrieved September 2026. Original work by Gege Akutami, Shueisha.

Once you know a slash is a plane, the question of how many pieces a body ends up in becomes pure combinatorics, and it has a clean closed-form answer.

Place n planes through a solid, arranged so that no two are parallel and no four meet at a point. The greatest number of regions they can produce is

$$r(n) = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} = \frac{n^3 + 5n + 6}{6}.$$

One slash makes 2 pieces, two make 4, three make 8. So far it looks like doubling, and here is where the intuition breaks: four slashes make 15, not 16. Five make 26. Six make 42.

The growth is cubic, not exponential, but cubic is quite enough. Ten slashes produce 176 fragments and twenty produce 1,351. The general version says n cuts in d dimensions give about $n^d/d!$ regions, so the efficiency of cutting is set by the dimension of the space you are cutting up.

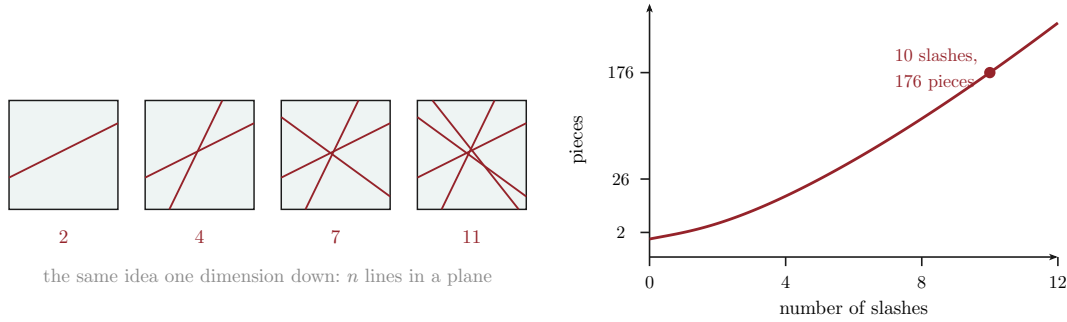


Figure 5: Left: the two-dimensional version, easier to draw and following the same law, where n lines cut a region into $\binom{n}{0} + \binom{n}{1} + \binom{n}{2}$ pieces. Right: the three-dimensional count $(n^3 + 5n + 6)/6$, which is what applies to a body and a set of slashes. It is worth noticing where this does not go exponential. Each new plane can only be divided by the planes already present, so the returns are polynomial, and the exponent is the dimension of the space.

Spiderweb is the same mathematics used deliberately. Sukuna puts a hand on the ground and carves a web: some number of radial strands, some number of concentric rings. That is a planar graph, and Euler's formula for planar graphs, $V - E + F = 2$, counts its faces exactly.

With n radials and m rings there are $V = nm + 1$ vertices counting the centre, and $E = 2nm$ edges, since every radial contributes m segments and every ring contributes n arcs. So

$$F = E - V + 2 = 2nm - (nm + 1) + 2 = nm + 1,$$

and subtracting the unbounded outer face leaves exactly nm pieces of ground. Twelve strands and six rings give seventy-two slabs, all of them unsupported at once. Sukuna does not break the ground. He makes it into a set of pieces and lets it discover that it is no longer a floor.



Figure 6: Spiderweb, chapter 215. Radial strands crossed with concentric rings, which is a planar graph with $nm + 1$ faces, one of them the outside world. Euler's formula predicts the number of slabs before a single one has fallen. Image source: Jujutsu Kaisen Wiki (Fandom), *Shrine*, retrieved September 2026. Original work by Gege Akutami, Shueisha.

6 Why Cutting Is So Cheap

There is a reason Sukuna can casually bisect a building while other characters strain against much smaller targets, and it has nothing to do with cursed energy.

Breaking something in bulk, pulverising it or blowing it apart, means doing work throughout its interior. That cost scales with volume, as the cube of the size. Cutting means separating material along a surface, and the cost scales with area, as the square. The ratio between them grows linearly with the size of the target, so the bigger the thing, the more absurdly the advantage tilts toward whoever is cutting.

Put numbers on it. The energy needed to part a material along a surface is its energy release rate G , about 10^4 joules per square metre for structural steel, which follows from a fracture toughness of $50 \text{ MPa} \cdot \sqrt{\text{m}}$ and a stiffness of 200 GPa . Take a tower fifty metres across and assume, generously, that three percent of any horizontal cross-section is load-bearing structure rather than air, glass and partition. That is 75 square metres of steel, so the cut costs

$$75 \text{ m}^2 \times 10^4 \text{ J/m}^2 \approx 7.5 \times 10^5 \text{ J},$$

about the chemical energy in two hundred grams of TNT. Now weigh that against what happens next. A hundred thousand tonne building whose centre of mass falls a hundred metres releases

$$mgh \approx 10^8 \times 9.81 \times 100 \approx 10^{11} \text{ J.}$$

The collapse is something like a hundred and thirty thousand times larger than the cut that caused it. Sukuna pays for a surface, and gravity pays for the building.

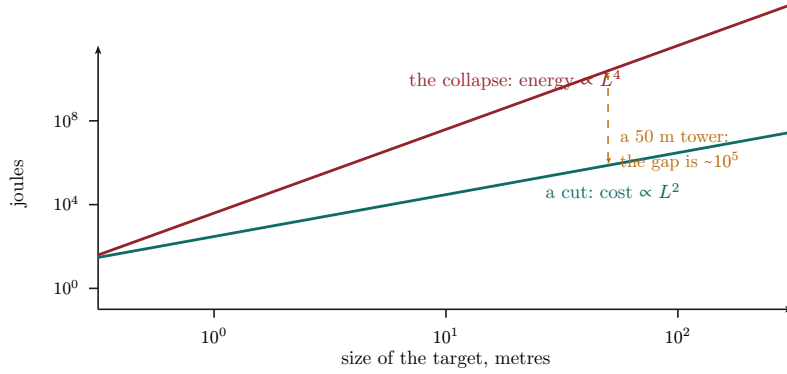


Figure 7: Why a cut is the cheapest way to destroy anything large. Separating material costs area and grows as L^2 ; the gravitational energy stored in the structure grows far faster. The two curves diverge without limit, so the advantage of cutting over brute force is not fixed, it increases with the size of the target. Against a building the gap is already about five orders of magnitude, which is the whole reason a single slash can drop a tower.

7 Cleave, a Cut That Reads Its Target

Dismantle is a fixed operation. Sukuna throws a plane, and whether it separates anything depends on how the target compares with the energy he happened to put behind it. That is why a weak Dismantle leaves gashes: the plane failed to reach all the way through, and by the previous section a cut that does not go all the way through does not raise b_0 at all.

Cleave removes that failure mode. Canon says it adjusts its power according to the target's toughness and cursed energy so as to cut them down in one attack, and Spiderweb does the same against the ground. In other words Cleave is not a fixed operation but a function that takes a property of the target as an input.

The distinction is worth stating carefully, because it is the difference between an attack and an algorithm. Dismantle applies a constant. Cleave measures first and then applies whatever exceeds what it measured. Against a target of toughness τ the required output is some $f(\tau)$, and Cleave supplies it by construction rather than by Sukuna guessing correctly.

That explains the one restriction canon puts on it. Outside his domain, Cleave needs physical contact. A function cannot evaluate an argument it has not been given, and touching the target is how the argument gets read. Inside Malevolent Shrine the contact requirement vanishes, which makes sense: the domain is a guaranteed hit, and a guaranteed hit is precisely a promise that the technique already has everything it needs about everything inside.

8 The Slash That Cuts the World

Now chapter 236, and the move that required none of the above.

Everything so far has been about cutting sets that live inside space. A body, a building, the ground: all of them are subsets $A \subset M$ of an ambient space M , and a slash separates A while leaving M completely untouched. Space does not notice. This is why Gojo was safe from all of it. Infinity governs what can approach him through M , and Dismantle is a thing approaching through M .

What Sukuna did was change which object the operation applies to. The manga says he broadened the target area to include space, existence and the world. In symbols, the escalation is two lines:

$$\text{Dismantle: } A \subset M \mapsto A_1 \sqcup A_2 \subset M, \quad M \text{ unchanged}$$

$$\text{World-cutting: } M \mapsto M_1 \sqcup M_2$$

To see why the second line is a different kind of statement, you need the definition of the thing being cut. A space M is an n -dimensional manifold if every point of it has a neighbourhood that looks like ordinary $\mathbb{R}^n$: zoom in far enough anywhere and you should find a plain patch of n -dimensional space with nothing strange about it. That single requirement is what licenses everything else. It is why you can put coordinates on M , why you can do calculus on it, and why distance means anything.

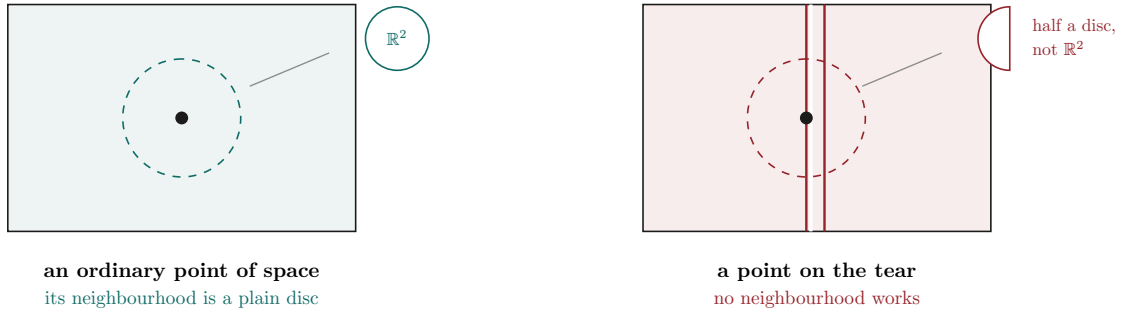


Figure 8: The definition that the world-cutting slash breaks. A manifold is a space in which every point, examined closely enough, has a neighbourhood indistinguishable from ordinary flat space. On the left that test passes. On the right it fails for every point along the cut, no matter how far you zoom in: the neighbourhood is half a disc with an edge, and an edge is a thing ordinary space does not have. Everything built on top of the manifold, coordinates, calculus, distance, is built on the left-hand picture being true.

9 Why Infinity Had No Answer

Infinity is a statement in differential geometry. In flat space the distance you travel along a path is $\int |d\mathbf{x}|$, and in a curved or distorted space it becomes

$$L[\gamma] = \int_{\gamma} \sqrt{g_{ij} \dot{x}^i \dot{x}^j} dt,$$

where the metric tensor g_{ij} is what converts a coordinate step into an actual distance. Gojo's technique inflates g_{ij} without bound as you approach him, so the integral diverges: every path that ends at Gojo has infinite length, and nothing finishes traversing it. That is the whole of Infinity, and against anything that has to arrive it is perfect.

Look at what the formula silently assumes. It needs a path γ that runs from the attacker to Gojo. It needs g_{ij} to be defined at every point along it. And underneath both, it needs M to be a manifold, because a metric tensor is a piece of structure laid on top of one.

The world-cutting slash does not fail any of those tests. It removes the thing the tests are about. Once M has been severed, there is no path from one side to the other, so there is no integral to evaluate and no length to inflate. The distance from the attacker to Gojo is not enormous. It is not a number.

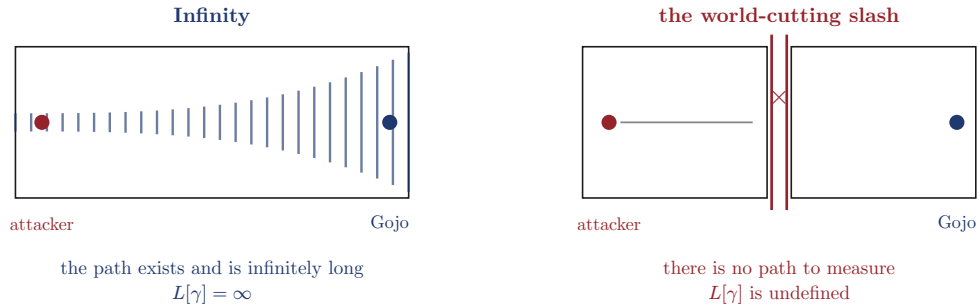


Figure 9: The difference that killed Gojo. On the left, Infinity leaves the route intact and makes it infinitely long, which defeats anything obliged to travel it. On the right, the route is gone. These are not two sizes of the same problem. Infinity is a function whose argument is a distance, and an operation performed on the manifold does not hand it a distance to work with. You cannot apply a function to something outside its domain.

There is a sharper way to say all of this, and it happens to close a loop with the mathematics usually applied to Gojo.

The standard analysis of Infinity observes that the halfway points an attacker must cross, $\frac{1}{2}$, $\frac{3}{4}$, $\frac{7}{8}$ and so on, form a countable set, and that every countable set has Lebesgue measure zero. The barrier occupies no length at all. From that one concludes, correctly, that as a physical obstruction Infinity is flimsy, and that its real strength has to come from the distorted metric rather than from the points themselves.

Now apply the same test to Sukuna's cut. A surface in three-dimensional space also has measure zero. It has no volume whatsoever. By the measure-theoretic criterion the two are identical, both negligible, both nothing.

And yet one of them stops nothing and the other stops everything, permanently. The property that separates them is codimension, from section 4, and codimension is a topological invariant. Measure cannot see it. Lebesgue measure assigns zero to a point, to a line and to a plane in three-space without distinguishing between them, which is exactly the distinction on which the entire outcome of the fight turns.

Gojo's defence is a measure-zero set that cannot disconnect anything. Sukuna's slash is a measure-zero set that disconnects completely. Both are nothing, by volume. Only one of them is a wall.

10 Conclusion

Sukuna's three cuts are three different answers to the question of what is being cut.

Dismantle throws a plane at a subset of space. Because a plane has codimension 1 it can separate, and because separation is combinatorial rather than brute, a handful of them produces a great many pieces: $(n^3 + 5n + 6)/6$ of them, 176 for ten slashes. Cleave throws the same plane but measures the target first and supplies whatever exceeds it, which is why it needs contact outside a domain and why it stops failing the way a weak Dismantle fails. Spiderweb is the same operation applied to the ground, and Euler's formula counts the slabs it makes before any of them has moved. All three are cheap for the same reason: a cut pays for area while destruction pays for volume, so against a fifty metre tower the slash costs about 10^5 times less than the collapse it triggers.

The last one is not an escalation of any of that. It is a change of object. Dismantle acts on $A \subset M$; the world-cutting slash acts on M . Infinity is a device for inflating the length of a path, and it is undefeated against everything that has a path. Sukuna's answer was to stop having one.

The limits are worth stating rather than leaving implied. Space, existence and the world is a phrase in a weekly manga and not a definition, and I have been treating it as though it meant a specific mathematical operation because that is the only way to get anywhere. The energy figures depend on a guess that three percent of a tower's cross-section is load-bearing, which is defensible to within a factor of a few and no better. Cursed energy still obeys no conservation law.

And there is a genuine physical objection to the whole premise, which I would rather raise myself than have raised for me. In classical general relativity you cannot do what Sukuna does. Robert Geroch proved in 1967 that a spacetime whose spatial topology changes must contain either closed timelike curves or a singularity; the manifold is not permitted to tear quietly. Whether quantum gravity relaxes this is an open question, and has been since Wheeler suggested that spacetime might be foamy at the smallest scales. So the world-cutting slash is not merely a very strong attack. It is the specific operation that the best theory of space we have says is forbidden, and forbidden for reasons of causality rather than of energy.

Which is a better compliment than the manga was reaching for. Every other technique in the series is a large number. This one is a change of category, and the mathematics has a name for what it does to the thing it is applied to. Gojo built the finest defence anyone has ever built against things that come toward you. He did not build one against an attack that declines to come toward him, because until chapter 236 nobody knew that was a direction you could decline to travel in.

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